NAPE AICE 2026 · TRACK 2 · TEAM TRINERGY · BUILT FOR HEIRS ENERGIES

# Intelligent Well Archive

Ask a question about decades of drilling reports. Get the answer, the page it came from, and an honest "not in the archive" when it is not.

CODE FINDS THE FACTS | AI PHRASES THEM | A VERIFIER CHECKS | ABSTAINING IS VALID

Flow and architecture pack for the three presenters (AI engineer, product designer, petroleum engineer). Data: Equinor Volve dataset. Palette: web/app/src/styles/tokens.css. No Heirs logo is used.

OVERVIEW

# The problem, the idea, the four principles

## The problem

Well history sits in daily drilling reports. Engineers search by hand, and nobody can easily show where a number came from. The task: answer 32 questions about Volve daily reports with traceable evidence (file, page, section), and say so when the archive cannot answer.

In this contest a correct abstention scores a full point. An invented citation scores zero.

## The idea

Turn every report into small records with stable IDs. Let code pick the right records. Let a small model write one short answer from only those records. Let code check the answer before it is accepted.

If the archive has no such field, the system says what it searched and stops.

1

### Code finds facts

Parser, router and context builder pick the evidence. The model never searches the archive.

2

### AI phrases

A small model (Haiku 5.5) turns the chosen lines into a short answer in a fixed JSON shape.

3

### Verifier checks

13 code checks. A number, date or ID that is not in the cited records is rejected.

4

### Abstention is valid

No field, no report, no answer. The evidence column says what was searched.

Source: CLAUDE.md (goal and scoring rules); submission/archive_qa/prompts.py; submission/archive_qa/verify.py; web/app/src/views/Impact.tsx (problem statement).

OVERVIEW

# Headline numbers

**24 + 8**
of 32 questions: 24 answered, 8 abstained (2 of them because the report does not exist)

**$0.022**
per question: 130 cached model calls cost $0.651 for all 32

**~14 s**
from a fresh kernel to a regenerated predictions.csv, offline, no API key, same file every time

**0**
invented citations: every cited ID must exist in the store and in the model's context

**1,783**
records from 32 reports, 8 wellbores, 54 PDF pages (1,707 archive records and 76 derived notes)

**31 → 40**
of 40 unseen questions, before and after the generalisation fixes. Tuned on those 40 (see page 21)

**345 + 99**
tests pass: 345 pipeline tests (1 skipped) and 99 API tests, from a fresh run

**13**
verifier checks, all in code. Confidence is set by rules, never by the model

Honest status: the engineer's hand check of the 32 answers and an independent test are **pending**. The key file `submission/eval/key_template.csv` has no verdicts yet. The hidden-test score is unknown.

Sources: GET /questions and GET /costs (web/server/API.md); submission/indexes/cost_summary.csv; submission/predictions.csv; web/server/data/MANIFEST.json; submission/DATA_NOTES.md; pytest runs on 2026-11-08 (root: 345 passed, 1 skipped; web/server/tests: 99 passed).

THE FLOW

# End to end, in one picture

## From question to cited answer

Code finds the facts. The model only phrases them. A verifier checks every answer. Abstaining is a valid answer.

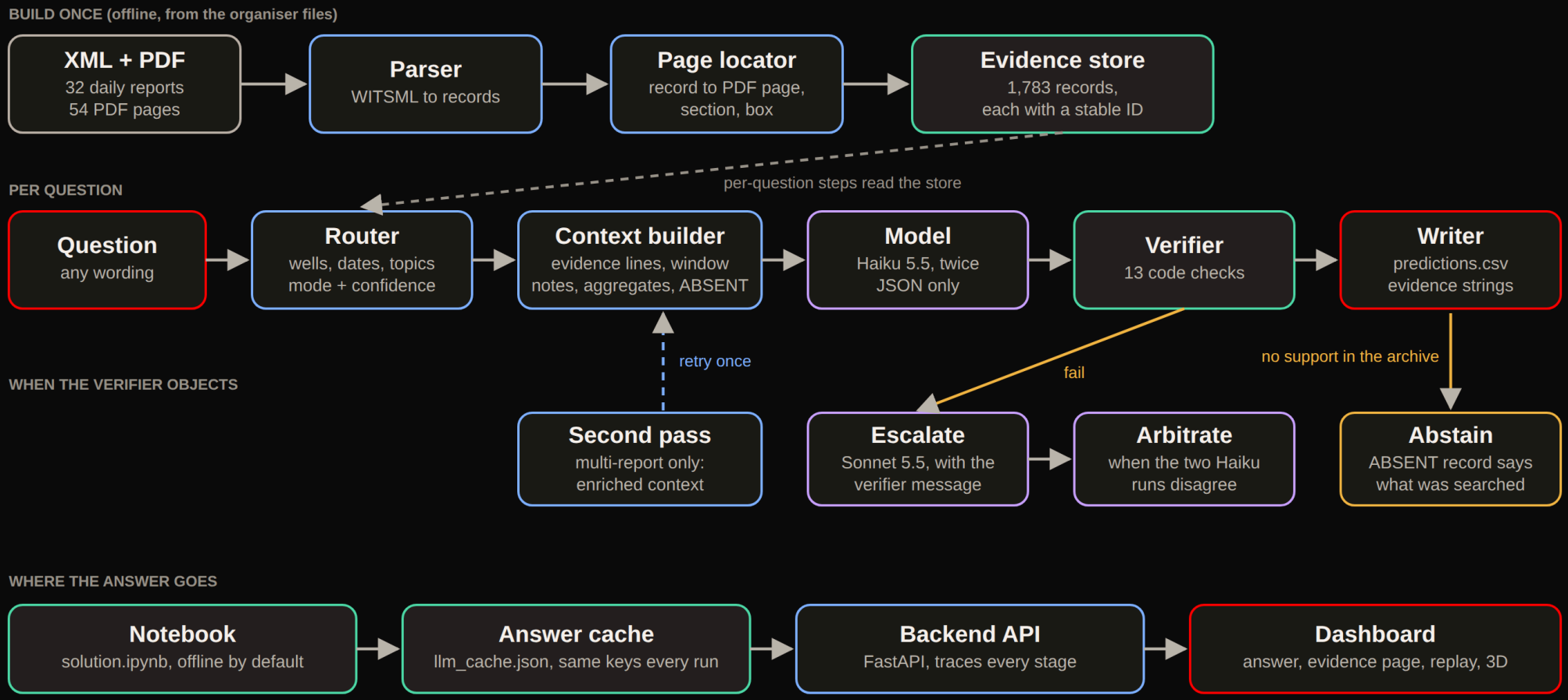



Diagram: docs/presentation/diagrams/pipeline.svg. Code: submission/archive_qa (answer.py, route.py, context.py, verify.py, write.py); web/server (API and trace).

STAGE 1 · DATA AND FORMATS

# What goes in: 32 reports, three formats

| FORMAT | WHAT IT IS | ROLE |
|---|---|---|
| XML | WITSML 1.4 drillReport, 32 files | Source of truth for values |
| PDF | The same report, 54 pages, all with a text layer (no OCR) | Page and section for citations |
| HTML | No text that is not also in the PDF | Not used |

**Wellbores (8):** 15/9-19-A, F-1, F-1-B, F-1-C, F-11-B, F-11-T2, F-12, F-15-D. There is no report for F-14.

A report dated D covers D-1 00:00 to D 00:00.

## What it prevents

- Reading a value from the wrong source: the XML wins when the PDF disagrees.
- Counting wells from a bad hint column: the manifest's `well_hint` leaks the year, so it gives 11 "wells". We count from file names (8).
- Encoding errors: 28 files declare ISO-8859-1, four declare nothing. Parsing raw bytes reads both.

## What comes out

A clean list of reports per wellbore and date, ready for the parser.

Missing values: `-999.99` is a sentinel (174 occurrences) and is never output as a value. The text "None" in 5 summary or forecast fields is also missing.

Source: submission/DATA_NOTES.md (Archive shape, Dates, Missing values); submission/archive_qa/wells.py (report discovery).

STAGE 2 · PARSER AND EVIDENCE STORE

# From XML to records with stable IDs

## Record IDs

```
F1C-20140326-FLUID-01
19A-19800101-STATUS-MDKICKOFF
F15D-20150810-ACT-01
F1C-20161022-FCST-FORECAST24HR
```

Well code, report date, record type, then a field name or a number. The same ID appears in the context, in the model's citation and in the evidence column.

## Record types (1,783)

| | | |
|---|---|---|
| status 619 | activity 370 | wellbore 275 |
| header 224 | survey 80 | note 76 |
| fluid 40 | summary 32 | forecast 32 |
| casing 9 | equip. failure 8 | pore pressure 7 |
| gas 3 | log, bit, strat, lithology 2 each | |

## What it does

- Reads each XML section into small records: one value, its unit, its time stamp, its report window.
- Keeps missing values as missing, with the original text.
- Builds **derived NOTE records** when XML and PDF disagree (64 flag fields, 11 days-behind, 1 bit end depth).
- Keeps a bit with two nozzle blocks complete. A parser bug once let the second block overwrite the first; it is fixed and tested.

## What it prevents

- Guessing: every later step names a record, not a paragraph.
- Silent loss of data from repeated tags.

Sources: submission/archive_qa/parse.py, store.py; records.parquet value counts (1,783 records; web/server/data/MANIFEST.json); submission/DATA_NOTES.md (XML vs PDF audit).

STAGE 3 · PAGE LOCATOR

# Every record gets a page, a section and a box

**99.2%**
table-row and narrative records located exactly

**94.5%**
of records the PDF prints

**78.8%**
of all records (the rest are XML only)

The locator searches each PDF page text, with all whitespace removed, because the PDF splits some words ("Summar y", "Sur vey Station"). It returns page, section heading and highlight rectangles.

## XML-only records

219 records sit in fields the PDF has no label for (for example `wellheadElevation`, `fixedRig`, hole-start fields). 64 are tightWell and HPHT, excluded by design because the PDF prints Y and the XML says 0 in all 32 reports. Their citations name the XML only, as "XML only: field; not printed in the PDF".

## What it prevents

- Citing a page that does not show the value.
- Showing a reviewer a PDF value that conflicts with the XML.

74 records are not located: 31 rig-code aliases, 32 NPD aliases, 11 daysBehind values (blank in the PDF) and 5 empty summary or forecast texts (heading only).

Sources: submission/DATA_NOTES.md (PDF location coverage); submission/archive_qa/locate.py, xmlonly.py; web/server/API.md (GET /pages, xml_only).

STAGE 4 · ROUTER

# From a question to the right reports

| MODE | WHEN |
|---|---|
| SINGLE_REPORT | one well and one date: the report dated D |
| MULTI_REPORT | several wells or dates, or a date range |
| ARCHIVE_WIDE | no well or date named: all 32 reports |
| EARLY ABSTAIN | the well or date is not in the archive: no model call, confidence 0.9 |

The router reads wells and dates in many spellings: F-11B, F 11 B, 15/9-F-11 B, NO 15/9-F-11 B, 15_9_F_11_B, F-1C and F-1-C; and 2014-03-14, 14 March 2014, March 14th 2014, 14/03/2014. Each date is paired with the well named for it ("the 24 September report for F-1B … the 2013-08-12 report for F-1").

## Rules that matter

- **Filter by well before date.** F-1-B and F-1-C both have a report dated 2014-02-22.
- **Exact date D first.** D+1 is used only as a recorded fallback, and the context says so.
- Topic words (cost, personnel, supervisor, weather, mud logging) set which sections to load.
- **Confidence level** in the trace: high = every target matched by exact date or range; medium = fallback or partial; low = whole archive. The API derives it from the router output.

## What it prevents

Sending the model the wrong well, the wrong day, or the whole archive when one report is enough.

Sources: submission/archive_qa/route.py (route_question, find_dates), wells.py (find_wells); tests/test_generalise.py (phrasings, pairing); web/server/API.md (Trace, router.confidence).

STAGE 5 · CONTEXT BUILDER

# What the model is allowed to see

```
[F1C-20140326-FLUID-01] fluid | density=1.03 g/cm3;
type=NaCl brine; locationSample=Active pit; md=4094 m
| report 2014-03-26 | at 2014-03-25T21:00:00
```

One line per record: ID, type, fields, report date, time stamp. A single-report context is small (T2-016: 4 lines, about 181 tokens). Multi-report contexts are capped at about 12,000 tokens, and recall forces in the records that match the question's distinctive words first.

### Special lines

- **WINDOW NOTE:** when a question names a clock time and a date, says the D-1 time stamp belongs to report D.
- **ABSENT record:** for cost, personnel, supervisor, weather, mud logging when the archive has no such field. It lists what was searched and near misses it rejected.
- **NOTE-DISCREPANCY:** shown only when the question mentions a field where XML and PDF disagree.
- **Aggregates:** for multi-report and archive-wide questions that ask for a count, extreme or comparison, code computes count, min, max and the source records. Aggregate lines get IDs like `AGG-00000000-WATERDEPTH`. A citation to one is replaced by its source records.
- **Archive index:** `IDX-00000000-WELLBORES` lists wellbores, reports per wellbore and date ranges.
- **ABSENT-search:** if none of the question's content words occurs in any field name or record text, an ABSENT record says what was searched.

## What it prevents

The model counting by hand across 32 reports, or reading a D-1 time stamp as a date mismatch. Code does the counting and the model reads the result.

Sources: submission/archive_qa/context.py (build_context, window_note), enrich.py (aggregates, ABSENT-search); GET /questions/T2-016 trace (est_tokens 181, 4 lines); submission/DATA_NOTES.md.

STAGE 6 · THE MODEL CONTRACT

# A short prompt, a strict shape, fixed settings

```
Answer the question using ONLY the evidence lines.
Reply with JSON only:
{"status":"answered"|"partial"|"abstain",
 "answer":str,"cited_ids":[str],"rationale":str}
Rules: cite the IDs of the lines you used; never invent
an ID; include units; if the lines do not contain the
information, status "abstain" with empty answer; if only
part of the question is supported, status "partial" and
answer that part only; never substitute a different
field for the one asked. rationale: one short sentence.
Record IDs carry the report date; a sample/event
timestamp may fall on the day before the report date
and still belongs to that report ...
```

About 250 tokens. The prompt's SHA is part of every cache key. Editing it invalidates the cache on purpose.

| SETTING | VALUE |
|---|---|
| Primary model | anthropic/claude-haiku-5.5, temperature 0 |
| Escalation model | anthropic/claude-sonnet-5.5, reasoning minimal |
| Output cap | 400 tokens (3,000 for multi-report); retry at 1,500 if cut off |
| Format | strict json_schema, four fields |
| Provider | pinned to Anthropic, no fallbacks, require_parameters off |
| Runs | two Haiku runs per question; the second only feeds a disagreement flag |
| Concurrency, budget | 8 workers, $3.00 spend guard per run |

## What it prevents

Free-form answers that cannot be checked, and the model deciding its own confidence.

Sources: submission/archive_qa/prompts.py; submission/archive_qa/config.py; submission/indexes/llm_cache.json (_meta: models, settings, prompt_sha 1628ad33a872932a); .env.example.

STAGE 7 · THE VERIFIER

# Thirteen checks, all in code

| # | CHECK | IN PLAIN WORDS |
|---|---|---|
| 1 | valid_json | The reply parses and has the four keys |
| 2 | status_valid | answered, partial or abstain |
| 3 | answer_nonempty | An answer has text |
| 4 | citations_well_formed | cited_ids is a list of strings |
| 5 | citations_in_context | Every cited ID was shown to the model |
| 6 | citation_required | An answer cites at least one record |
| 7 | abstain_citations | An abstention cites only ABSENT records |

| # | CHECK | IN PLAIN WORDS |
|---|---|---|
| 8 | numbers_supported | Every number is in a cited record |
| 9 | units_match | The unit next to a number matches the record |
| 10 | dates_times_supported | Every date and time is in a cited record, its window or the question |
| 11 | no_false_not_found | No "nothing found" for a well that has a matching record |
| 12 | parts_addressed | Each part of a multi-part question is answered |
| 13 | no_missed_evidence | An abstention is not contradicted by a matching line |

Details: dates and times are whole tokens, not digits. A derived number (sum, difference, mean) passes only if the answer says it is derived and both operands are cited and stated. Decimal commas, list numbers "(1)", short IDs like "ACT-13" and "24:00" are handled so correct answers are not rejected.

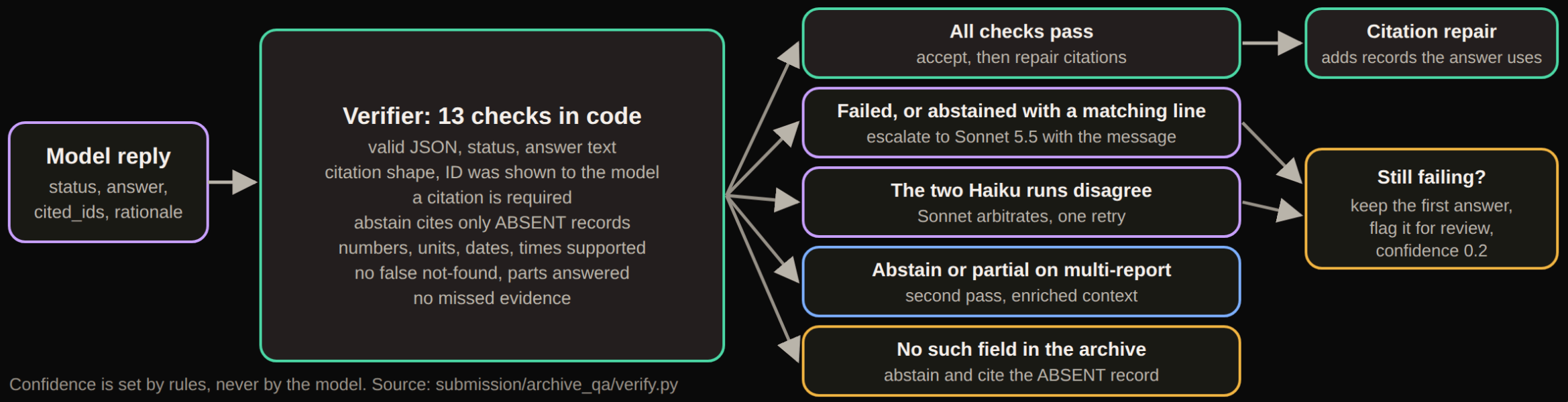
**What it prevents:** a wrong number that sounds right, a citation to a record the model never saw, and an abstention when the line was in the context. Failures go back to the model as feedback. They are never silently fixed.

Sources: submission/archive_qa/verify.py (verify, module docstring); check names from GET /questions/T2-017 trace (verifier[].checks).

STAGE 8 · ESCALATION, ARBITRATION, SECOND PASS

# What happens when the first answer is not good enough

## The trust layer: nothing reaches predictions.csv unchecked




Confidence is set by rules, never by the model. Source: submission/archive_qa/verify.py

**Escalate.** Verifier failed, or the model abstained although a matching line exists: one Sonnet 5.5 call that sees the verifier's message. Never escalates for a plain "no field" abstention.

**Arbitrate.** The two Haiku runs disagree on status or key numbers: Sonnet decides. If it fails the checks, one retry with the message, then run 1 is kept and flagged.

**Second pass.** Multi-report or archive-wide only, once: the first pass abstained, was partial or tripped the abstain flag. Retry with aggregates, a field catalogue and full records of the named wells.

**Citation repair.** Adds records whose content the answer clearly uses, and checks the answer still verifies. A partial answer is escalated once.

Sources: submission/archive_qa/answer.py (_answer_question), repair.py; diagram docs/presentation/diagrams/trust_layer.svg. Second pass real-data test: session run, not committed [VERIFY]: 5 of 5 abstain or partial cases rescued.

STAGE 9 · ANSWER CACHE AND WRITER

# Same answers every run, with or without a key

## Answer cache

Key = sha256 of system prompt, context, question text, feedback, model, settings, prompt SHA and run label (length-prefixed). The file `submission/indexes/llm_cache.json` holds 130 entries (119 KB) and records the models and settings, so an offline replay rebuilds the same keys with no configuration.

| SITUATION | RESULT |
|---|---|
| Cached, any key state | answer from the cache, no network |
| Not cached, key present | model called; new answer saved to the runtime cache only |
| Not cached, no key | row "NOT EVALUATED: no API key and no cached answer", confidence 0, empty evidence |

## Writer

Writes `predictions.csv` (question_id, answer, evidence, confidence, notes). Evidence is built **only from records in the store**: unknown IDs are dropped. Per report, PDF page references first, then XML IDs; XML-only values are cited as such; abstentions list the files searched.

| CONFIDENCE RULE | VALUE |
|---|---|
| One verified record (single report) | 0.95 |
| Two or three records | 0.80 |
| Four or more records | 0.70 |
| Partial | 0.40 to 0.60 |
| Abstain, field not in archive | 0.85 |
| Abstain, report does not exist | 0.90 |
| Verifier failed | 0.20 |

Sources: submission/archive_qa/llm.py (cache_key, LayeredCache), write.py, verify.py (confidence); submission/indexes/llm_cache.json; submission/README.md (hybrid mode).

WORKED EXAMPLE 1 · T2-016 · THE DATE WINDOW

# A sample stamped the day before the report

*What was the density of the NaCl brine sampled from the active pit on well 15/9-F-1-C on 2014-03-26?*

| STAGE | WHAT HAPPENED |
|---|---|
| Router | single_report, confidence high: exact date, one report `15_9_F_1_C_2014_03_26` |
| Context | 4 lines, about 181 tokens: summary, forecast, one fluid record |
| The trap | the sample is stamped `2014-03-25T21:00:00`, one day before the question's date, but inside the report that covers D-1 to D |
| Model | two Haiku runs, both answered 1.03 g/cm3 citing `F1C-20140326-FLUID-01` |
| Verifier | all checks pass; no escalation |

## Result

**1.03 g/cm3**, confidence 0.95, verified

`15_9_F_1_C_2014_03_26.pdf p.1 (Drilling Fluid); 15_9_F_1_C_2014_03_26.xml F1C-20140326-FLUID-01`

Cost when generated: $0.0003 (2 calls, 1,946 prompt tokens). Replayed from the cache for free.

## Why it matters

In the benchmark, date-window cases scored 12 percent across eight models. Larger models abstained more here. Haiku answered. When a question also names a clock time (T2-004: 18:00 on 2013-06-28), a WINDOW NOTE is added to the context.

Sources: GET /questions/T2-016 (trace); submission/predictions.csv; submission/DATA_NOTES.md (Dates); tools/llm_bench/RECOMMENDATION.md (local benchmark notes, not committed).

WORKED EXAMPLE 2 · T2-031 · ABSTENTION

# The failure is logged. The cost is not in the archive.

*What was the cost of the replacement parts for the equipment failure logged on well 15/9-F-15-D on 2015-08-10?*

| STAGE | WHAT HAPPENED |
|---|---|
| Router | single_report, confidence high: `15_9_F_15_D_2015_08_10` |
| Context | 25 lines, about 1,760 tokens. The activity records do say "equipment failure" and "lost communication with tractor". An `ABSENT-cost` record says: searched 1 report, 56 records, no cost field exists. |
| Model | two Haiku runs both abstained |
| Citations | set in code: an abstention may cite only ABSENT records |
| Verifier | passes; no escalation (abstaining on a missing field is correct) |

## Result

**Not answerable from the archive.** No cost field exists (searched 1 report, 56 records).

`15_9_F_15_D_2015_08_10.pdf: all sections checked, no cost field`

Confidence 0.85. Cost when generated: $0.0009.

## Why it matters

The archive has no cost, AFE, personnel, supervisor or weather fields in any XML or PDF. A correct abstention scores a full point and an invented number scores zero. The ABSENT record is what a reviewer reads as proof that we looked.

Sources: GET /questions/T2-031 (trace); submission/predictions.csv; submission/DATA_NOTES.md (Fields the archive does not have); submission/archive_qa/context.py (_absent_record).

WORKED EXAMPLE 3 · T2-017 · MULTI-REPORT

# A verifier failure, an escalation and a citation repair

*During the October 2016 plug and abandonment campaign, in what order did the rig move between wells while cutting and pulling production risers, from the 15/9-F-15-D report on 2016-10-19 to the 15/9-F-1-C report on 2016-10-22?*

| STAGE | WHAT HAPPENED |
|---|---|
| Router | multi_report: range 2016-10-19..2016-10-22 spans wells, so 4 reports (F-15-D, F-12, F-11-B, F-1-C) |
| Context | 93 lines, about 6,003 tokens; recall forced in 16 matching records |
| Haiku run 1 | answered, citing 8 records. It also wrote "24" and "134.9" (runs differed on these) |
| Verifier | **failed**: "numbers in the answer not found in the cited records: ['24']" |
| Escalation | Sonnet 5.5 called with that message. Its answer **passed** all checks |
| Repair | citation repair added `F1C-20161022-ACT-01` and `ACT-02` |

## Result

Order: F-15-D, F-14, F-12, F-11 B, F-1 C, then planned moves to F-7 and F-9 A. Confidence 0.70 (ten records).

3 calls, 38,301 prompt tokens, $0.034 when generated. Sonnet cost 91 percent of that.

## Why it matters

The cheap model got the story right but added a number that no cited record contains. The verifier caught it. The escalation fixed it for a few cents. Nobody had to read the answer to know it was wrong.

Sources: GET /questions/T2-017 (trace: router, contexts, model_calls, verifier, decisions); submission/predictions.csv; web/server/samples/T2-017.json.

WORKED EXAMPLE 4 · T2-020 · ARCHIVE-WIDE

# No well named, so the whole archive is searched

*On which completion days did the crew land the BOP onto the surface riser and make up the NT2 connector, and what BOP pressure tests were done on those days?*

| STAGE | WHAT HAPPENED |
|---|---|
| Router | archive_wide, confidence **low**: no well or date named, all 32 reports |
| Context | 442 candidate records, cap about 12,000 tokens: kept 117, dropped 325 (forced matches first). Confidence loses 0.05 for truncation |
| Haiku runs | both answered but disagreed on numbers and times |
| Arbitration | Sonnet 5.5 arbitrated; its answer verified and was committed |
| Repair | added `F15D-20131214-ACT-14` |

## Result

Two days: 2013-06-27 (F-11-B report of 2013-06-28) and 2013-12-13 (F-15-D report of 2013-12-14), with the BOP pressure tests of each day. Confidence 0.65, seven cited records.

3 calls, 67,502 prompt tokens, $0.057 when generated.

## Why it matters

This is the hardest class. The context does not hold everything, so confidence is lower by rule and the trace shows what was dropped. For new counts and extremes, code now computes the answer first (aggregates). The 32 published contexts, including this one, are unchanged by that.

Sources: GET /questions/T2-020 (trace); submission/predictions.csv; submission/archive_qa/context.py (TRUNCATED note); verify.py (confidence).

DATA TRAPS

# Seven things that would have produced confident wrong answers

| TRAP | HOW IT IS HANDLED |
|---|---|
| **D-1 window.** Report D covers D-1 to D. | Every record carries its window; a WINDOW NOTE and the verifier accept the D-1 stamp. |
| **1980 placeholder.** The 15/9-19-A report dated 1980-01-01 has a 1997 spud date and no text. | Still cited: its kick-off 2178 m is printed on PDF page 1 (T2-001). |
| **Same-date reports.** F-1-B and F-1-C both have 2014-02-22. | Filter by well, then by date. |
| **Well-name variants.** "15/9-F-11 B", F-11B, F-11-T2; XML says "NO 15/9-F-1 C"; bare F-1 is the main bore. | One canonical name; tested spellings; each date paired with its own well. |

| TRAP | HOW IT IS HANDLED |
|---|---|
| **XML vs PDF.** PDF prints Y for tightWell and HPHT in all 32 reports; XML says 0. Days ahead or behind blank in 11 PDFs. Bit end depth 2574 vs 2577. | XML is authoritative; derived NOTE records; the locator never gives these fields a page. |
| **XML-only fields.** NPD numbers, rig code, wellhead elevation, fixedRig, mudClass, BOP rating, bit grading. | Cited as "XML only: field; not printed in the PDF". |
| **Western Atlas.** A company name in 19-A 1997-08-29. | It is a wireline logging company, not mud logging. T2-032 abstains and says so. |

Also: -999.99 and the text "None" are missing, never values. There is no report for F-14, and F-15-D has no report on 2013-12-15 (T2-027, T2-029 abstain with no model call).

Sources: submission/DATA_NOTES.md (all items); submission/archive_qa/discrepancy.py, xmlonly.py; submission/predictions.csv (T2-001, T2-027, T2-029, T2-032).

MODEL STRATEGY

# A small model does the work. A bigger one is called only when the code objects.

| MODEL (16 CASES) | ANSWER ACC. | $ PER CALL | P50 S |
|---|---|---|---|
| **claude-haiku-5.5** | **100%** | 0.00026 | 1.21 |
| claude-sonnet-5.5 | 94% | 0.00528 | 2.12 |
| claude-opus-5.5 | 94% | 0.00980 | 2.16 |
| gemini-3-flash-preview | 94% | 0.00108 | 1.34 |
| deepseek-v4-flash | 87.5% | 0.00012 | 1.28 |
| gpt-oss-120b | 87.5% | 0.00013 | 0.82 |
| qwen3.7-plus | 75% | 0.00062 | 2.70 |
| llama-4-scout | 56% | 0.00021 | 2.82 |

Caveat: 16 cases, one call each. One case is 6 points. Good for failure patterns and cost, not for ranking close models. Zero invented citations in 405 calls.

## Why Haiku is primary

Best accuracy on our cases, decision-stable on 4 of 4 repeats, 1.2 s median, about a twentieth of Sonnet's price. The task is reading a few lines, not reasoning, and the verifier does the checking.

## Where the money goes (32 questions)

Haiku: 104 calls, 80 percent of calls, 12.7 percent of cost ($0.083). Sonnet: 26 calls, 20 percent of calls, 87.3 percent of cost ($0.568). Total $0.651.

## Provider pinning

Anthropic requests must be pinned to the Anthropic provider with require_parameters off. With it on, every request returned HTTP 404. If the pin is unavailable on a key, the run falls back to default routing for the same model and logs a warning.

Sources: tools/llm_bench/RECOMMENDATION.md (local benchmark notes, not committed [VERIFY]); GET /costs (web/server/API.md); submission/archive_qa/llm.py (ProviderUnavailable fallback).

REPRODUCIBILITY

# Reviewers rerun the notebook in under a minute, with no key

| RUN | WHAT HAPPENS | TIME |
|---|---|---|
| No key, 32 published questions | all answers from the committed cache; predictions.csv identical (md5 90717d3e7767eee3f53b739bb32923fc) | about 14 s from a fresh kernel (about 10 s of compute) |
| Key, 40 new questions, cold cache | 81 live calls, valid output file, 0 questions needing review | about 21 s, $0.12 |
| Key, 3 demo questions, cold cache | 6 live calls, 2 answered, 1 abstained | about 7 s incl. kernel start, $0.003 |
| No key, uncached question | "NOT EVALUATED: no API key and no cached answer", never a guess | instant |

**Key names:** OPENROUTER_KEY first, then OPENROUTER_API_KEY, OpenRouter_Key, openrouter_api_key; environment or `.env`. The first notebook cell prints which one it found, never its value.

## Why it is deterministic

- Models are not bit-identical even at temperature 0. The cache is what fixes the answers.
- Cache keys include the question text, so a changed question misses instead of returning a stale answer.
- New answers go to a git-ignored runtime cache. The committed cache is never modified by a run.
- A different question file writes `predictions_<file>.csv`, never the committed predictions.csv.

## Tests (fresh run, 2026-11-08)

345 passed, 1 skipped (pipeline, about 31 s). 99 passed (API, about 97 s). Tests assert the 32 contexts and routes are unchanged, that no socket opens with no key, and that predictions.csv is byte-identical.

Sources: submission/README.md; tests/test_generalise.py; submission/archive_qa/llm.py; session runs of the notebook with NAPE_QUESTIONS_CSV (timings are one run each [VERIFY on your machine]); pytest output 2026-11-08.

GENERALISATION

# Forty questions the pipeline had not seen

| CATEGORY | BEFORE | AFTER |
|---|---|---|
| Single-report lookups (24) | 23 | 24 |
| Cross-report comparisons (4) | 0 | 4 |
| Archive-wide counts and maxima (5) | 1 | 5 |
| Fields the archive does not hold (7) | 7 | 7 |
| **Total (40)** | **31** | **40** |

Expected answers were computed by separate code reading the XML directly. Cost for the run: $0.276 before, $0.061 after.

## Honest caveat

We fixed the failures after seeing them, so 40 of 40 is not a clean estimate. A second 40 from a different random seed also scored 40 of 40, but it uses the same question templates, and that result is not saved in the repository [VERIFY]. The scorer is lenient on numbers.

## What we fixed

- **Router:** a date followed by "report for F-1" now pairs with F-1, not the earlier well. One question had routed to the wrong report.
- **Computed aggregates:** counts, extremes and comparisons are computed in code and shown as evidence lines.
- **Field-name section match:** a question about "date drilling was completed" now loads the section holding `dateDrillComplete`.
- **Generic absence:** an ABSENT-search record when no content word matches anything.
- **Second pass:** one retry with an enriched context. It did not fire on the final run. In a test with aggregates switched off it fired on 5 of 8 and rescued all 5; the other 3 were confident wrong answers, which it does not touch [VERIFY: session test, not committed].

Pending: the engineer's hand check of the 32 answers and an independent test set.

Sources: submission/eval/generalisation (gen.py, run.py, score.py, questions.csv, expected.json); GET /generalisation (baseline_mean_score 0.775, mean_score 1.0, cost_usd 0.060687); commit eecf7cd; submission/eval/key_template.csv (no verdicts).

SYSTEM ARCHITECTURE · BACKEND API

# The API wraps the frozen pipeline and records every stage

| ENDPOINT | RETURNS |
|---|---|
| `GET /health` | status, live mode, models, budget left |
| `GET /questions` | 72 questions (32 + 40) with status and flags |
| `GET /questions/{id}` | answer, evidence, cited records, full trace |
| `POST /ask, /ask/stream` | any question; the stream sends one event per real stage |
| `GET /pages/{record}.png` | the PDF page with the evidence highlighted |
| `GET /archive, /trajectories` | reports, wellbores, survey paths (approximate) |
| `GET /costs, /generalisation` | cost split, unseen-set results |

The server starts with a data pack (7.5 MB: records, 54 page images, boxes). No organiser raw file is needed at run time.

## Trace schema

**router** (mode, confidence, reports and why) · **contexts** (tokens, records, aggregates, ABSENT) · **model_calls** (role, model, cache hit, provider-reported tokens, latency, cost) · **verifier** (13 named checks) · **escalations**, **second_pass**, **arbitration** · **cited** · **decisions** · **totals** · **events**.

## Guardrails

- **Frozen pipeline:** the trace builder watches from outside; tests prove predictions and caches are byte-identical after live asks.
- **Clarification gate:** a vague question with no well, date or field term and no cached answer gets examples and costs nothing.
- **Limits:** 400-character cap, 6 per minute and 40 per day per client, 4 concurrent, 120 s timeout, spend cap $2.00 (persisted).
- **Key hygiene:** never returned or logged; errors pass through a redactor.

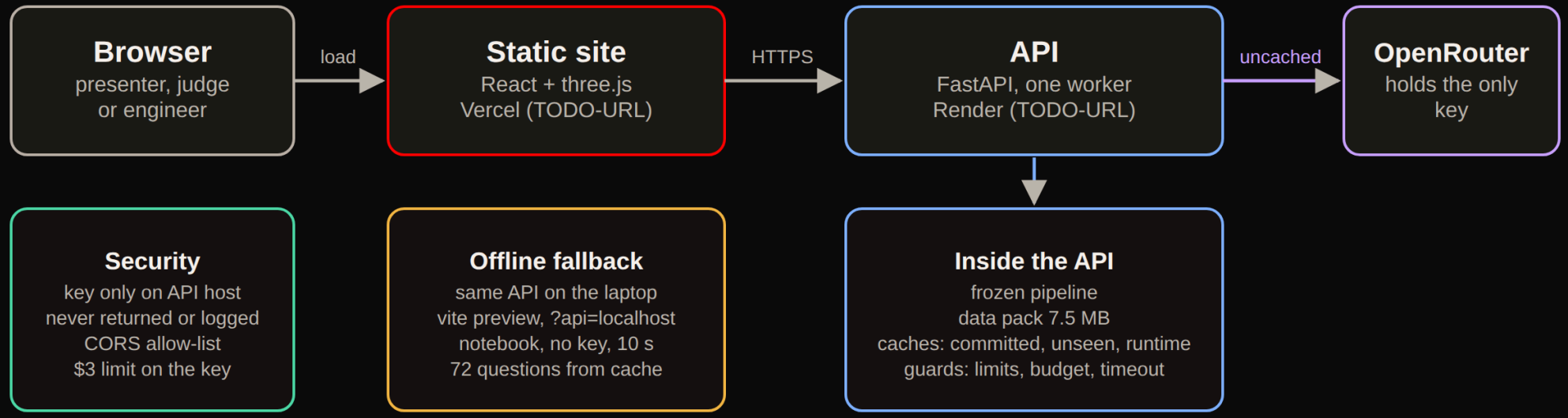
Sources: web/server/API.md (endpoints, Trace, Guarantees, Configuration); web/server/app.py, guards.py, safety.py; web/server/data/MANIFEST.json.

SYSTEM ARCHITECTURE · DASHBOARD AND DEPLOYMENT

# Eleven views, one real API, an offline fallback

## Deployment topology

The repository deploys nothing by itself. Hosted URLs are placeholders (TODO-URL). See web/DEPLOY.md.




Sources: web/DEPLOY.md, web/render.yaml, web/server/API.md, web/app/README.md

| VIEW | WHAT IT SHOWS |
|---|---|
| Overview, Questions | live counters; all 72 questions by set, mode, status |
| Answer | answer, confidence rule, PDF page with highlights, trace |
| Mission control | 2D replay of the real trace; 3D view (three.js), Focus mode |
| Ask | your own question, live trace, designed error states |

| VIEW | WHAT IT SHOWS |
|---|---|
| Generalisation, Cost, Archive | unseen set before and after; cost and cache; 32-report timeline |
| Impact, Present, Styleguide | use cases and roadmap for Heirs; guided tour; tokens |

Everything on screen comes from the API. Fonts, icons and libraries are bundled.

Sources: web/app/README.md (Views, Run, Run: hosted); web/DEPLOY.md; web/render.yaml; web/server/API.md (security headers); diagram docs/presentation/diagrams/deployment.svg.

COST, LIMITS, ROADMAP

# What it costs, what it cannot do, what comes next

## Cost

| ITEM | USD |
|---|---|
| 32 questions, cached (130 calls) | 0.651 |
| per question | 0.0217 |
| Escalation (26 calls) | 0.568 |
| Haiku calls (104) | 0.083 |
| 40 unseen, after fixes | 0.061 |
| 3 demo questions, live | 0.003 |
| Model benchmark [VERIFY] | 0.62 |

A typical new question costs well under a cent. A hard archive-wide one up to about five cents.

## Limits

- One field, 32 reports. No scanned pages here: every PDF page has a text layer. OCR is not built.
- No cost, personnel, supervisor or weather fields exist, so those questions abstain.
- Hand check by a petroleum engineer is pending. Independent test pending. Hidden-test score unknown.
- Confident wrong answers on multi-report questions are caught only if a number or ID breaks a check.
- Free Render plan sleeps; its spend counter resets on restart. The key's own limit always holds.
- Locator reaches 78.8% of records; the rest are cited from the XML.

## Roadmap

| | |
|---|---|
| TODAY | Volve prototype: cited answers, verifier, abstention |
| NEXT | Pilot on a small set of real Heirs daily reports, scored on Heirs engineers' own questions |
| THEN | Completion reports, well tests and surveys, with page evidence |
| LATER | One searchable, auditable memory across the asset base |

Not yet done: OCR for scans, a larger blind test, deployment inside Heirs, role-based access.

Sources: GET /costs (web/server/API.md); submission/indexes/cost_summary.csv; web/DEPLOY.md; web/app/src/views/Impact.tsx (roadmap); submission/DATA_NOTES.md; tools/llm_bench/RECOMMENDATION.md (local, not committed [VERIFY]).

ONE-PAGE CHEAT SHEET

# Numbers you may be asked, with their sources

| FACT | VALUE | SOURCE |
|---|---|---|
| Reports, wellbores, PDF pages | 32, 8, 54 | submission/DATA_NOTES.md |
| Records in the store | 1,783 (1,707 archive, 76 derived notes) | web/server/data/MANIFEST.json |
| Located in the PDF | 99.2% table and narrative, 94.5% of printed, 78.8% of all | submission/DATA_NOTES.md |
| 32 answers | 24 answered, 8 abstained (2 early) | GET /questions; submission/predictions.csv |
| Confidence values used | 0.65, 0.70, 0.80, 0.85, 0.90, 0.95 | submission/indexes/answers.json (local artefact) |
| Cached calls and cost | 130 calls, $0.651, $0.0217 per question, 949,726 prompt tokens | submission/indexes/cost_summary.csv |
| Model split | Haiku 104 calls, 12.7% of cost; Sonnet 26 calls, 87.3% | GET /costs |
| Models, settings | Haiku 5.5 and Sonnet 5.5, temperature 0, 400/3,000/1,500 token caps, Anthropic pin | submission/archive_qa/config.py |
| Unseen 40 | 31 to 40 correct; $0.276 to $0.061; tuned on them | GET /generalisation; commit eecf7cd |
| Offline rerun | about 14 s from a fresh kernel, md5 90717d3e7767eee3f53b739bb32923fc | submission/README.md; clean-clone run 2026-11-08 |
| Tests | 345 pipeline (1 skipped) and 99 API | pytest, 2026-11-08 |
| Verifier checks | 13 | GET /questions/{id} trace |
| API limits | 400 characters, 6 per minute, 40 per day, $2.00 server budget | web/server/API.md |
| Benchmark | 8 models, 16 cases, 405 calls, 0 invented citations | tools/llm_bench (local) [VERIFY] |

All values were read from the files named in this table on 2026-11-08. Items marked [VERIFY] are not in the committed repository.